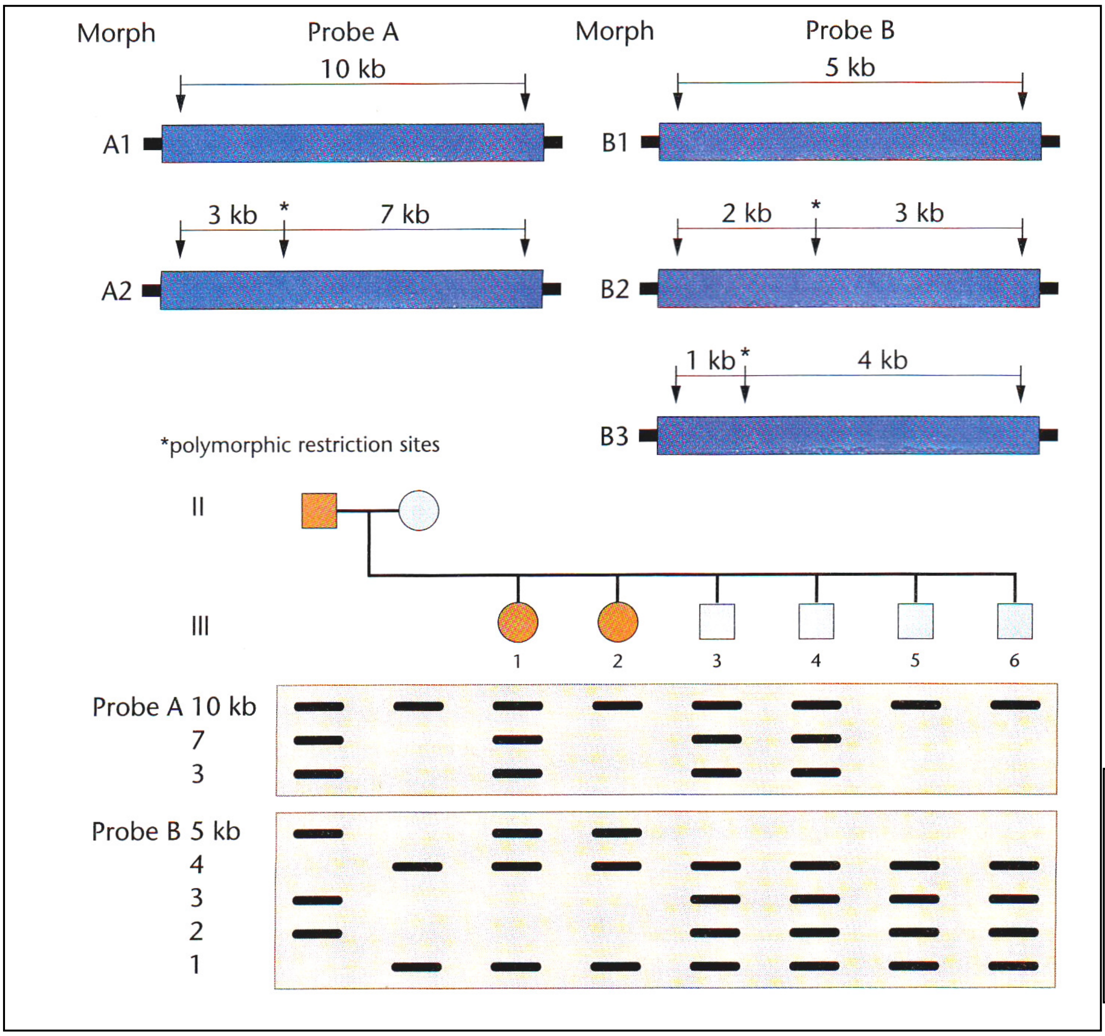
Answer **all** questions.

1. In a study of a rare genetic disease, DNA samples from two generations of an affected family were obtained and subjected to Restriction Fragment Length Polymorphism (RFLP) linkage analysis. One RFLP is found at locus **A** on chromosome 10 (identified by **probe A**) and the other is found at locus **B** on chromosome 21 (identified by **probe B**). The various alleles of the two RFLP loci are shown in Fig. 1.1. The results obtained from the RFLP analysis are shown in the same figure below.

Restriction fragment sizes detected



(a)

Fig. 1.1

(a) In the space provided in **Fig. 1.1**, indicate with

(i) the symbols (+) and (-), the relative positions of the electrodes, [1]

(ii) and an **arrow,** the direction of migration of restriction fragments during gel electrophoresis. [1]

(b) State what is meant when a certain locus is described as 'polymorphic'. [1]

(c) Describe how RFLP may be used to map the chromosomal locus of a dominant mutant allele. [3]

(d) Based on the results shown in **Fig. 1.1**,

(i) state the genotypes of individuals **II-2** and **III-1** with respect to the **A** and **B** loci. [2]

Individual **II-2** :

Individual **III-1** :

(ii) state on which chromosome the disease gene is located. [1]

(e) Individual **III-4** marries a woman with the genotype **B1B1.** They have a child with the genotype **B1B1**. The father suspects the child is illegitimate.

(i) Explain the genetic basis for this suspicion. [2]

(ii) If the child is not illegitimate, suggest one possible explanation how the child could have acquired the **B1B1** genotype. [2]

(iii) Suggest, with a reason, what further test(s) may be used to confirm the suspicion. [2]

[Total 15]

2. **(a)** Cystic fibrosis (CF) is caused by failure of a protein ion channel, the cystic fibrosis transmembrane conductance regulator (CFTR), which spans the plasma (cell surface) membrane.

Describe the inheritance of CF in humans. [4]

(b) Heterozygotes for the most common mutation causing CF may be more resistant to bacterial infections of the gut than homozygotes for the normal allele.

The bacterium, *Salmonella typhi*, which causes typhoid fever in humans, can infect mouse gut cells. Mouse gut epithelial cells were genetically engineered to express either the normal or the mutant allele of the human CF gene. The two types of cell were grown in tissue culture. Cultures of each cell type were separately incubated with three different strains of *S. typhi*.

The numbers of bacteria taken up by the cells are shown in Fig. 2.1.

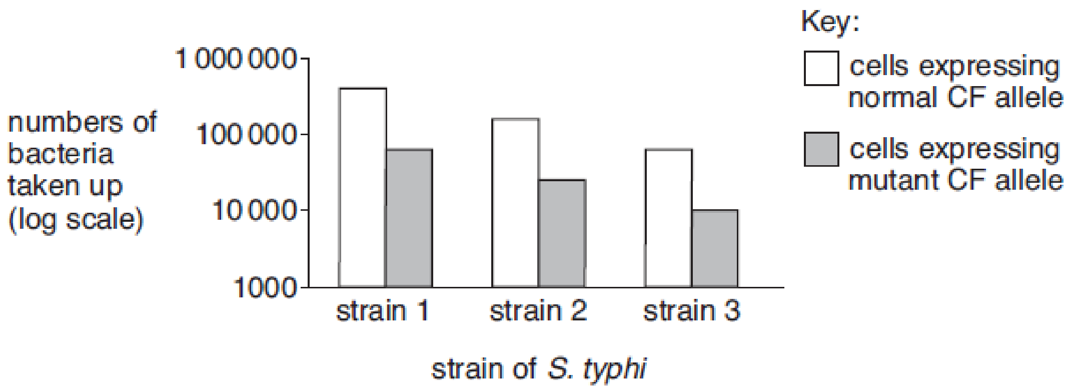


Fig. 2.1

With reference to Fig. 2.1,

(i) compare the effects of the normal and mutant alleles of the human CF gene on the number of bacteria taken up by mouse gut epithelial cells; [3]

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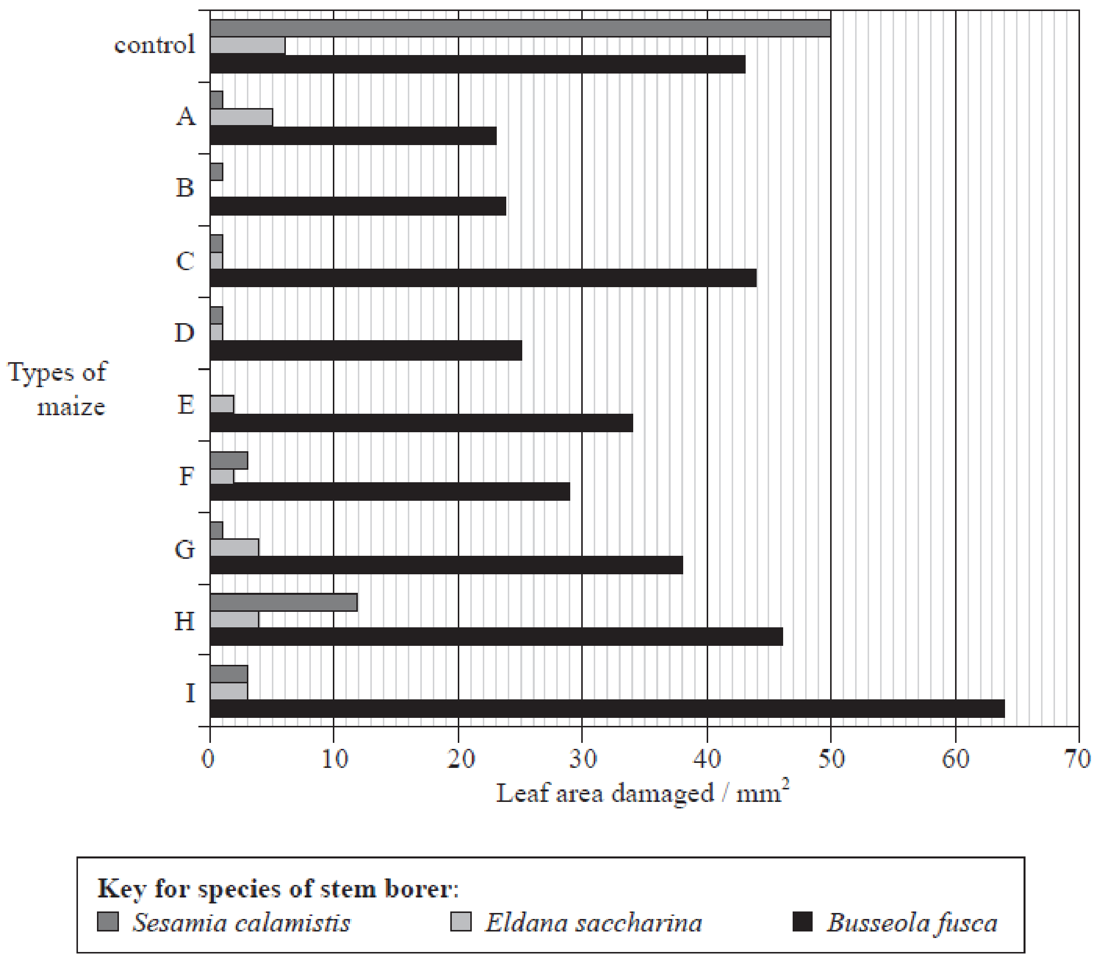
(ii) suggest how the **different** alleles of the human CF gene affect the number of bacteria taken up by mouse gut epithelial cells. [2]

[Total 9]

3. Genetic engineering allows genes for resistance to pest organisms to be inserted into various crop plants. Bacteria such as *Bacillus thuringiensis* (Bt) produce proteins that are highly toxic to specific pests.

Stem borers are insects that cause damage to maize crops. In Kenya, a study was carried out to see which types of Bt genes and their protein products would be most efficient against three species of stem borer. The stem borers were allowed to feed on nine types of maize (**A**–**I**), modified with Bt genes. The graph below shows the leaf areas damaged by the stem borers after feeding on maize leaves for five days.



[Source: S Mugo et al., Figure 3 from "Developing Bt maize for resource-poor farmers—Recent advances in the IRMA project", *African Journal of Biotechnology* (2005), volume 4, number 13, pp. 1490-1504]

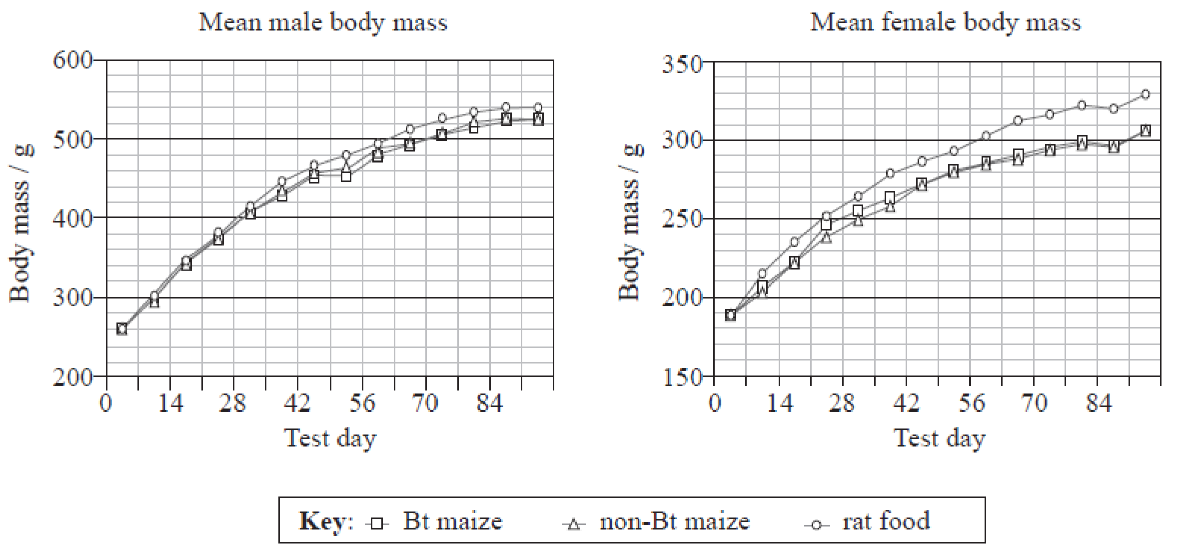
(a) Calculate the percentage difference in leaf area damaged by *Sesamia calamistis* between the control and maize type H. Show your working. [2]

(b) Discuss which species of stem borer was most successfully controlled by the genetic engineering of the maize plants. [3]

Before the use of genetically modified maize as a food source, risk assessment must be carried out. A 90-day study was carried out in which adult male and female rats were fed either:

- seeds from a Bt maize variety
- seeds from the original non-Bt maize variety
- commercially prepared rat food.

All the diets had similar nutritional qualities.



[Source: Reprinted from Linda A. Malley _et al_., "Subchronic feeding study of DAS-59122-7 maize grain in Sprague-Dawley rats", _Food and Chemical Toxicology_, volume 45, issue 7, pp. 1277-1292., Copyright (2007), with permission from Elsevier]

(c) Evaluate the use of Bt maize as a food source on the growth of the rats. [2]

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(d) Comment on the use of Bt maize as a food source compared to the other diets tested. [1]

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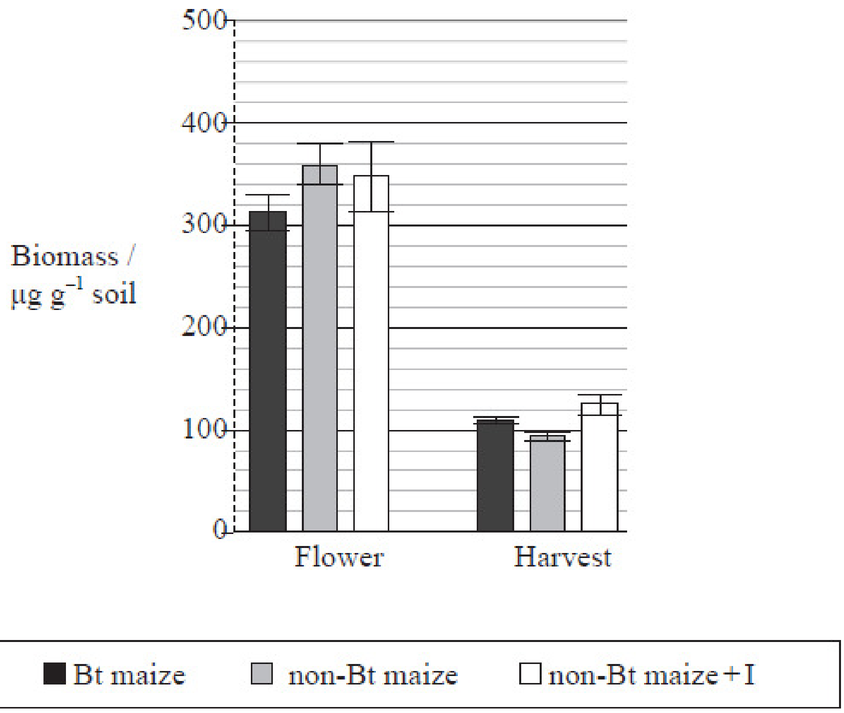
Studies have shown that Bt proteins are released by plant roots and remain in the soil. One study looked at the biomass of microorganisms in soil surrounding the roots of:

- Bt maize
- non-Bt maize
- non-Bt maize with an insecticide (**I**).

The graph below shows the biomass of microorganisms at two different times in the growth cycle of the plants (Flower and Harvest). Error bars represent standard error of the mean.

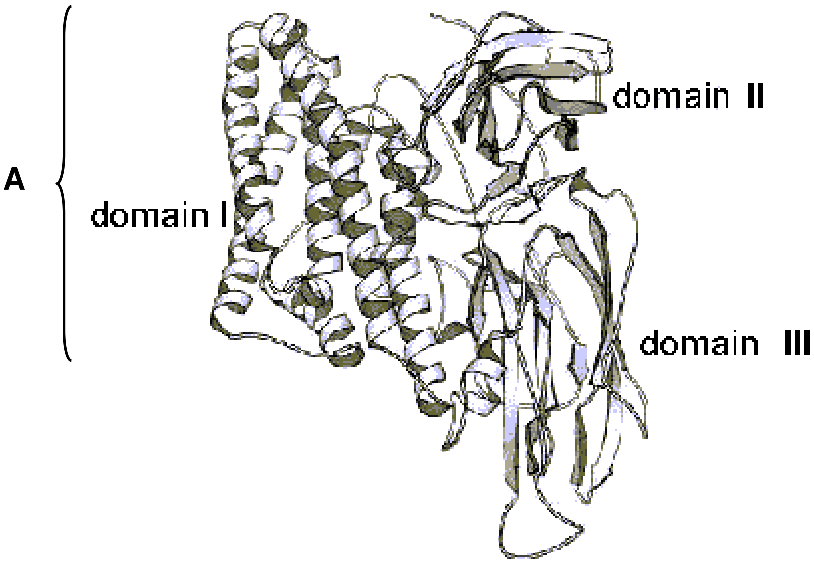


[Source: Reprinted from M. Devare et al., "Neither transgenic Bt maize (MON863) nor tefluthrin insecticide adversely affect soil microbial activity or biomass: A 3-year field analysis", *Soil Biology and Biochemistry*, Volume 39, Issue 6, pp. 2038-2047, © (2007) with permission from Elsevier]

(e) Compare the biomass of microbes in the soils surrounding the roots of Bt maize and non-Bt maize. [2]

(f) The researchers' original hypothesis stated that microorganisms would be negatively affected by the Bt protein released by the plant roots. Discuss whether the data supports the hypothesis. [2]

Bt proteins act as toxins to insects, primarily by destroying epithelial cells in the insect's digestive system. Below is the three-dimensional structure of one such protein.



(g) (i) Outline how structure **A** is held together. [2]

(ii) Region **A** is inserted into the membrane. Deduce, with a reason, the nature of the amino acids that would be expected to be found in this region. [2]

[Total 16]

4. Skill A – Planning

You are required to plan, but not carry out, an investigation into the effect of light intensity on the rate of light dependent reaction of photosynthesis in a leaf extract using the dye DCPIP. When DCPIP is reduced, it changes from blue colour to colourless.

DCPIP (blue) + electrons → reduced DCPIP (colourless)

A leaf extract can be made by mixing finely ground leaf with cold buffer solution. This lead extract should be kept cold and in the dark except when taking samples.

When capillary tubes are dipped into this leaf extract, the solution rises up into the capillary tube and remains there.

The leaf extract should be placed into the capillary tubes to conduct the experiment.

Your planning must be based on the assumption that you have been provided with the following equipments and materials which you must use:

- capillary tube
- fresh spinach leaves
- sharp knife
- mortar and pestle
- cold buffer solution
- ice
- a light-proof aluminium foil
- DCPIP solution (freshly made)
- lamp (chosen so it does not generate heat)
- ruler
- fine sharp sand
- a room which can be made dark
- cloths
- a variety of different sized beakers, measuring cylinders, syringes and pipettes for measuring volumes

Your plan should: have a clear and helpful structure to include

- an explanation of theory to support your practical procedure
- a description of the method used including the scientific reasoning behind the method
- an explanation of the dependent and independent variables involved
- relevant, clearly labelled diagrams
- how you will record your results and ensure they are as accurate and reliable as possible
- proposed layout of results tables and graphs with clear headings and labels
- the correct use of technical and scientific terms.

[Total: 12]

Free-response question

Write your answer to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labelled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

5 **(a)** Using a **named example**, describe the characteristics of adult stem cells. [8]

(b) Explain why embryonic stem cells are preferred over adult stem cells for the treatment of genetic diseases. [6]

(c) Describe three differences between the Polymerase Chain Reaction (PCR) and DNA replication. [6]